Scalability & Future Plan

Date	06 July 2026
Team ID	SWTID-2026-4324
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited dataset for model training	May reduce prediction accuracy	High
2	Supports limited crop recommendations	Restricts usability for different regions	Medium
3	No real-time IoT sensor integration	Cannot provide live field monitoring	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited users	Deploy on cloud with load balancing for more users.
2	Data Storage	Uses local database	Upgrade to cloud database for larger storage.
3	Performance	Suitable for small datasets	Optimize code and ML model for faster predictions.
4	Security	Basic user authentication	Implement advanced authentication and data encryption.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve crop prediction accuracy using larger datasets	3 Months	More accurate recommendations
Phase 2	Integrate weather API and real-time data	6 Months	Better farming decisions
Phase 3	Develop Android mobile application	9 Months	Increased accessibility for farmers
Phase 4	Add IoT sensor integration and cloud deployment	12 Months	Real-time monitoring and scalable system